Week 10
Network Security and wrap up

IN 1011 Operating Systems

Network Security

- Introduction to Networks
- Network Security
- Encryption
- Further Topics
- Wrap up

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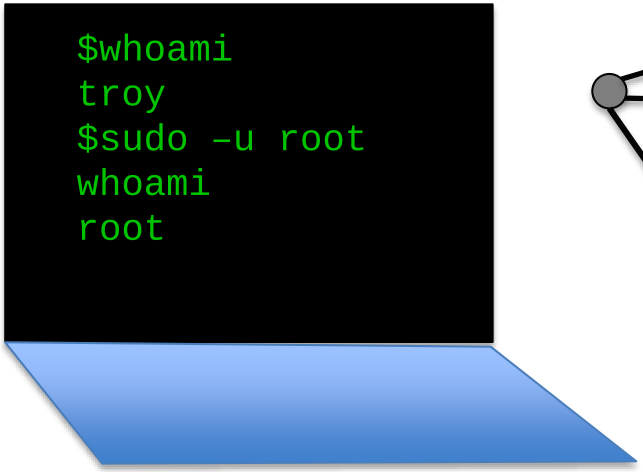
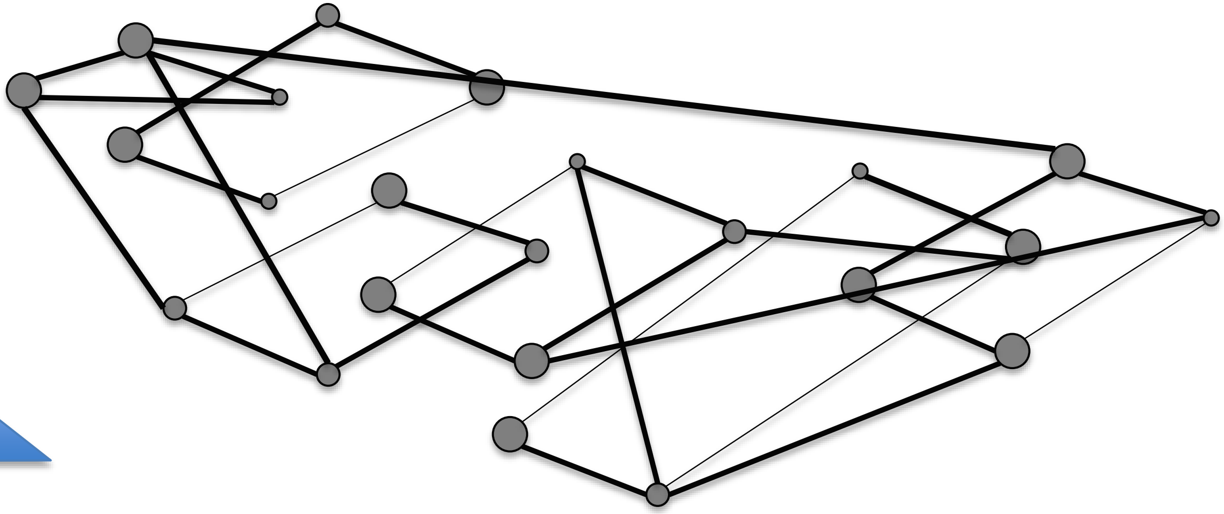
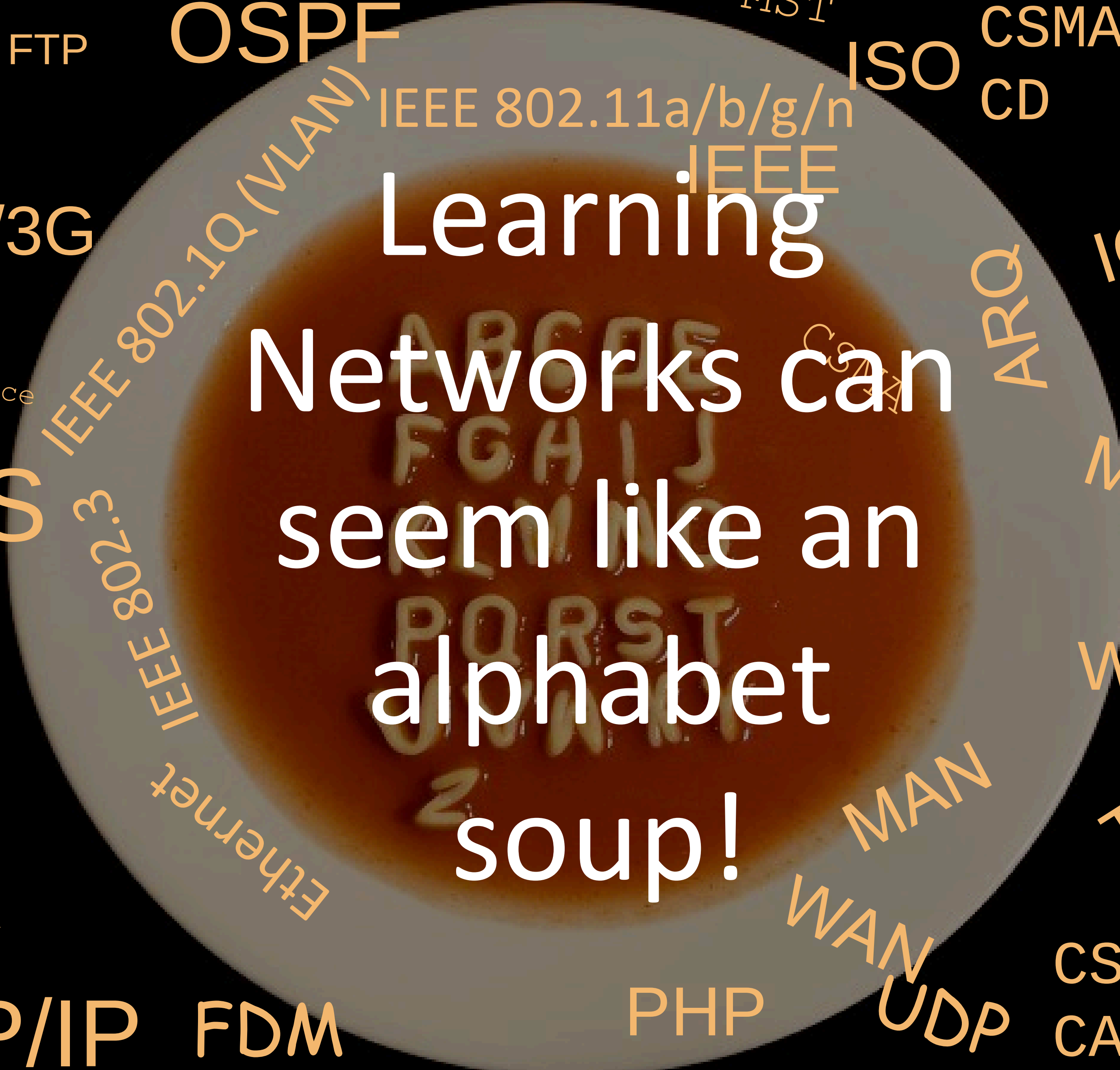
Networks and distributed systems

- We mainly dealt with operating systems for a stand **alone** computer
- But computers are **internetworked** and must communicate and cooperate with each other.





Learning Networks can seem like an alphabet soup!



PAN FTP OSPF MST ISO CSMA-CD ITU
HTTP IEEE 802.11a/b/g/n IEEE
2.5G/3G IEEE 802.1Q (VLAN) ICANN
p-persistence CSMA ARQ IPv.6
DNS IEEE 802.3 MIME
CSMA IEEE WWW
LAN Ethernet MAN TDM
S-ALOHA WAN
TCP/IP FDM PHP UDP CSMA-CA

Networks made easy

Networks allow the *exchange of information*

Networks made easy

Networks allow the *exchange of information* between *hosts*

Networks made easy

Networks allow the *exchange of information* between *hosts* through a series of series of *applications*

Networks made easy

Networks allow the *exchange of information* between *hosts* through a series of series of *applications* that are ruled by a series of *protocols*.

Networks made easy

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Networks made easy

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Networks made easy

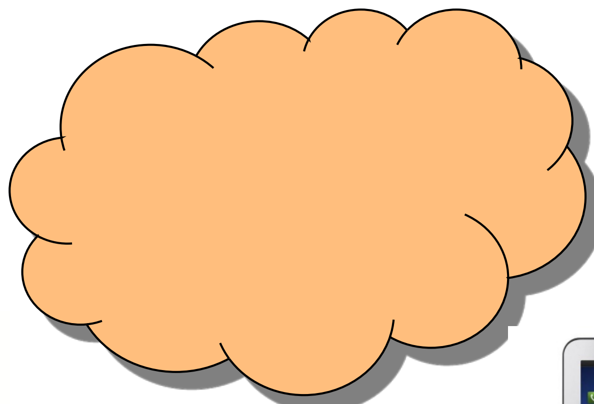
Networks allow the *exchange of information* between *hosts* through a series of series of *applications* that are ruled by a series of *protocols*. The information exchanged by the hosts together with other *control messages* are transmitted through a large variety of *access media* that interconnect many intermediate *devices*.

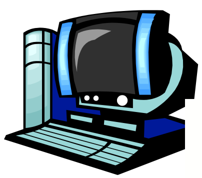
The Internet: 1 – Devices = hosts

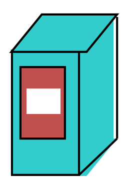
- A series of very different devices connected to “The Internet”

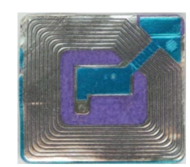










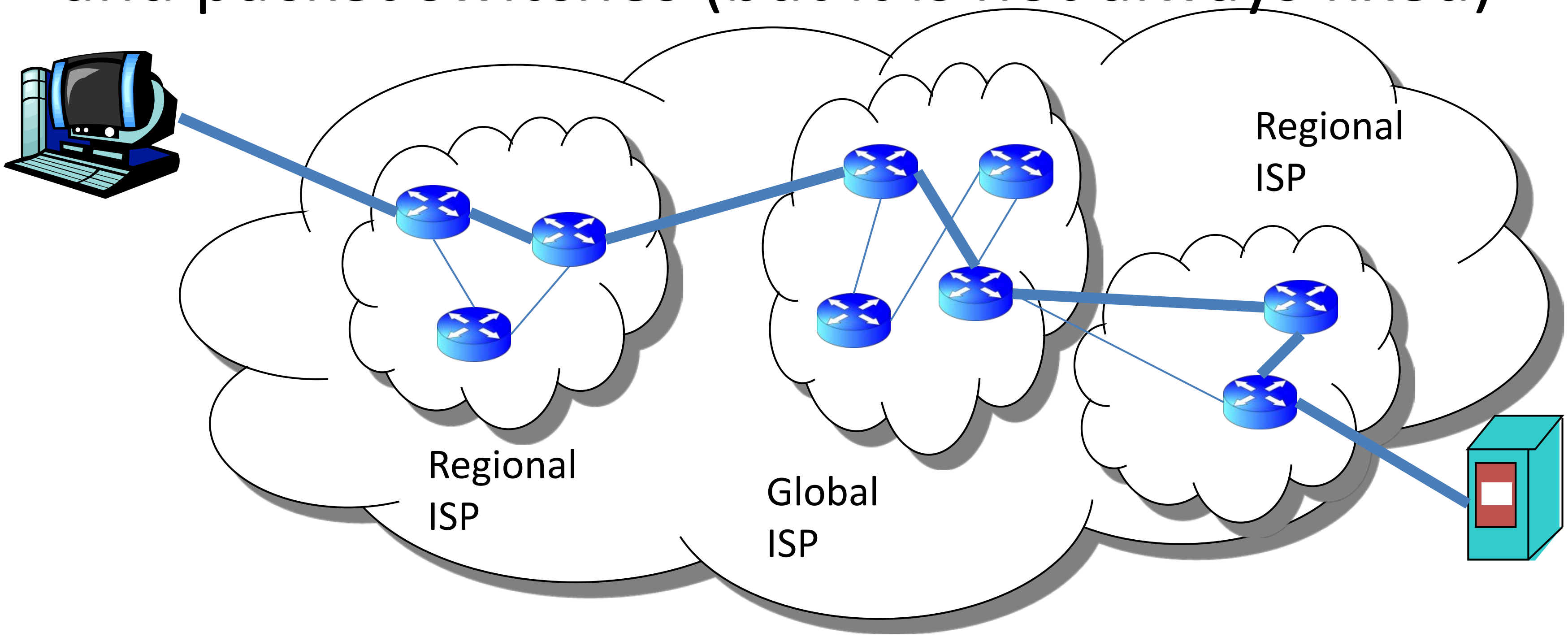



The Internet: 2 – Access and Media

- Each device, sometimes called hosts or end systems, needs to “connect” through some media

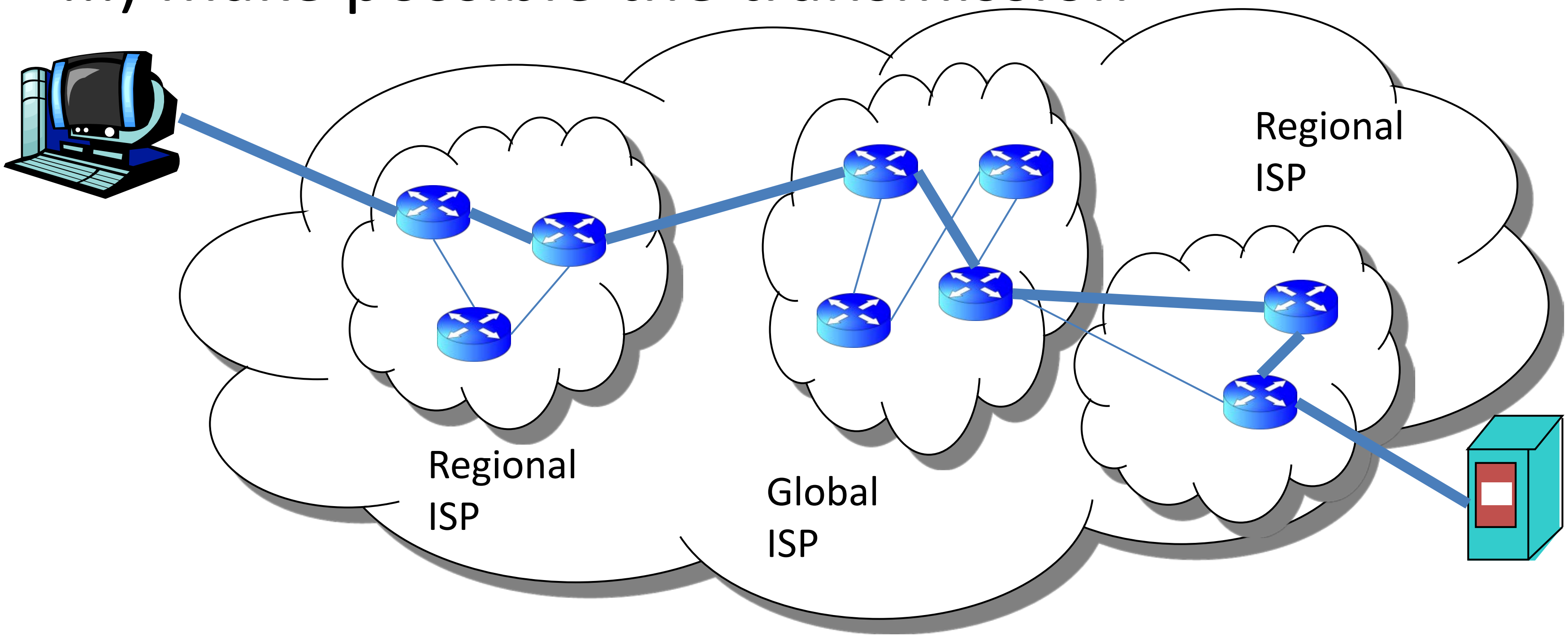
The Internet: 3 - Routes

- Once physically connected, a route between devices is established through a series of links and packet switches (but it is not always fixed)



The Internet: 4 - Protocols

- The information is transported through packets and a series of protocols (http, ftp, smtp, tcp, ip, ...) make possible the transmission

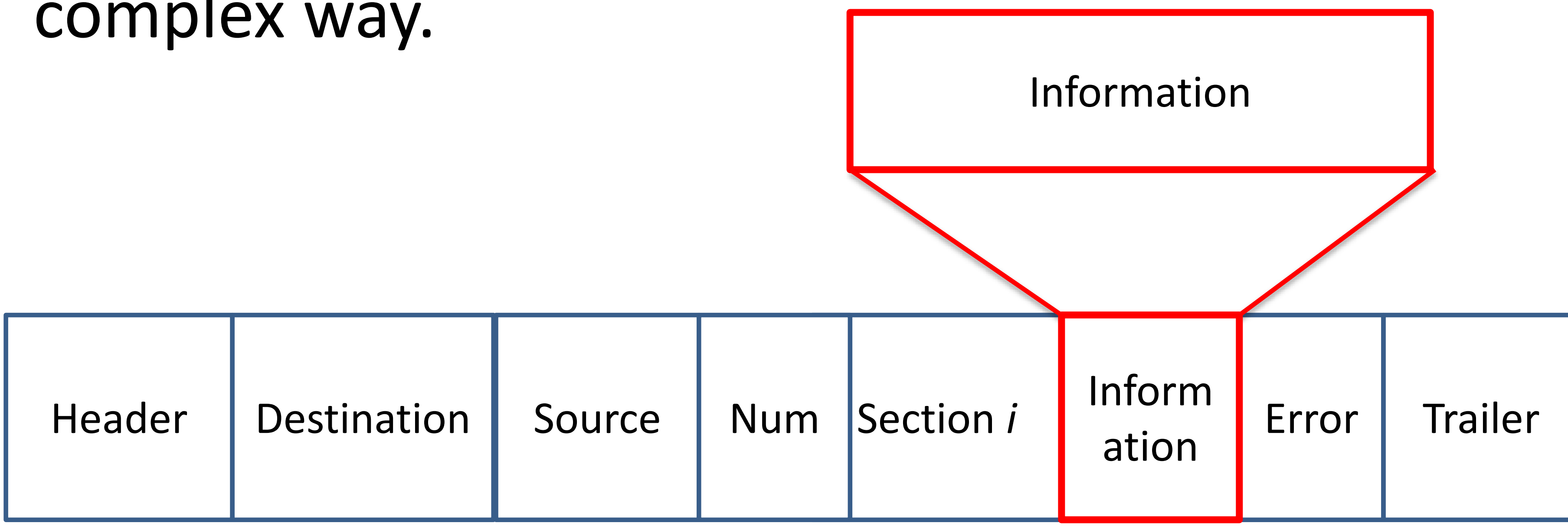


The Internet: 5 - Applications

- Alternatively, the internet is an infrastructure that provides services to applications

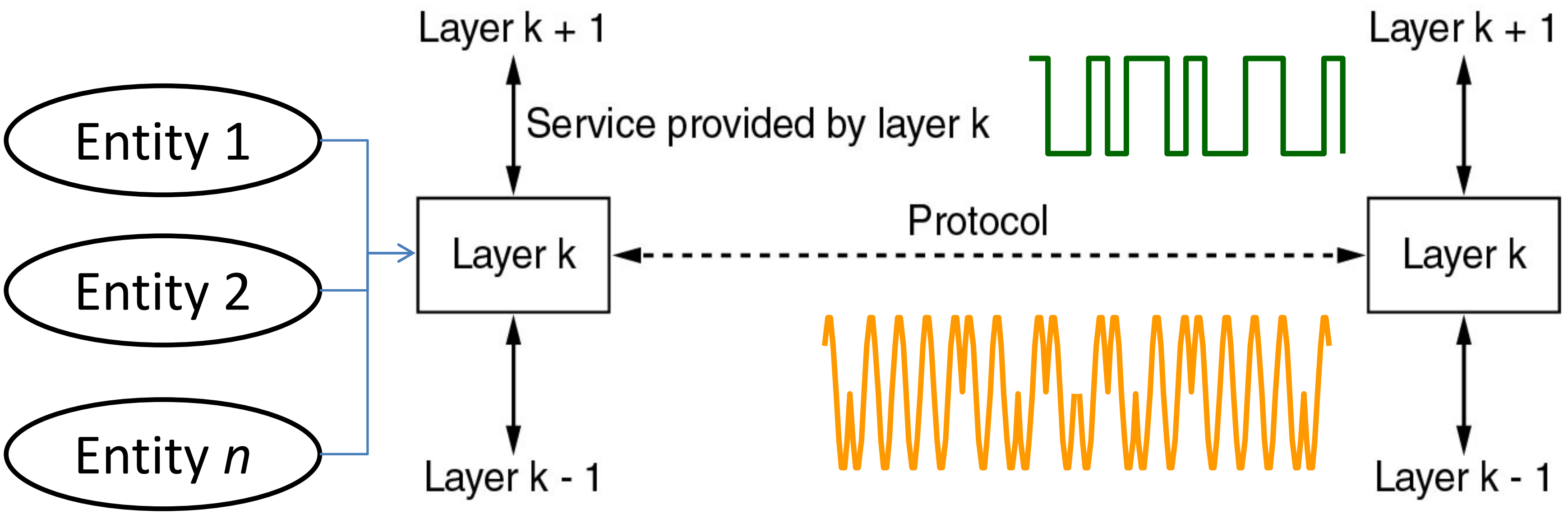
The Internet: 6 – Information and Messages

- In the end, all that matters is the exchange of information, but **the information** is embedded with a lot of ***other information*** in a rather complex way.



Layers, Services, Interfaces, Entities, Protocols, Models, *etc.*

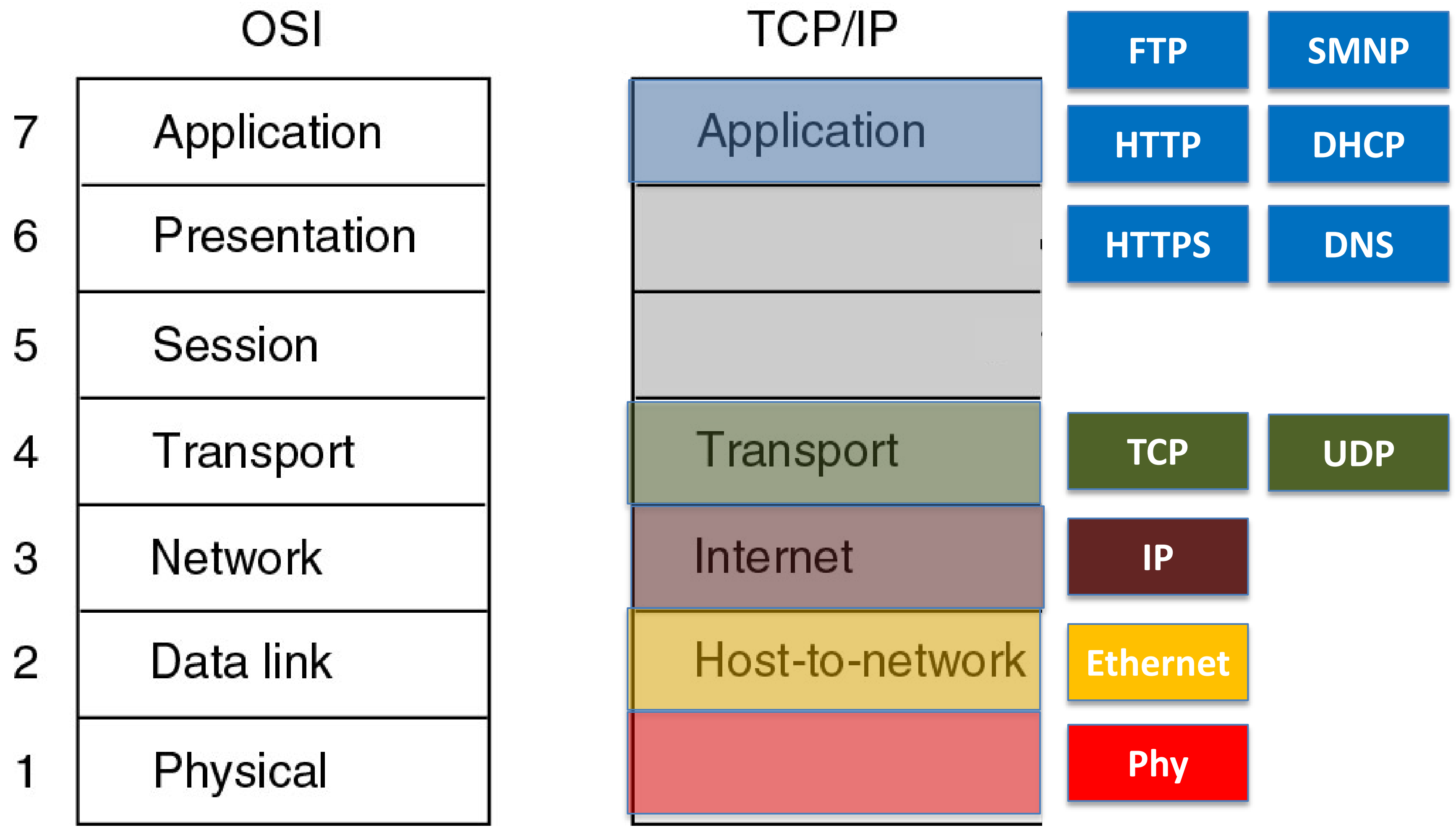
- A set of layers and protocols is called a network architecture. The definition of layers form a **model.**



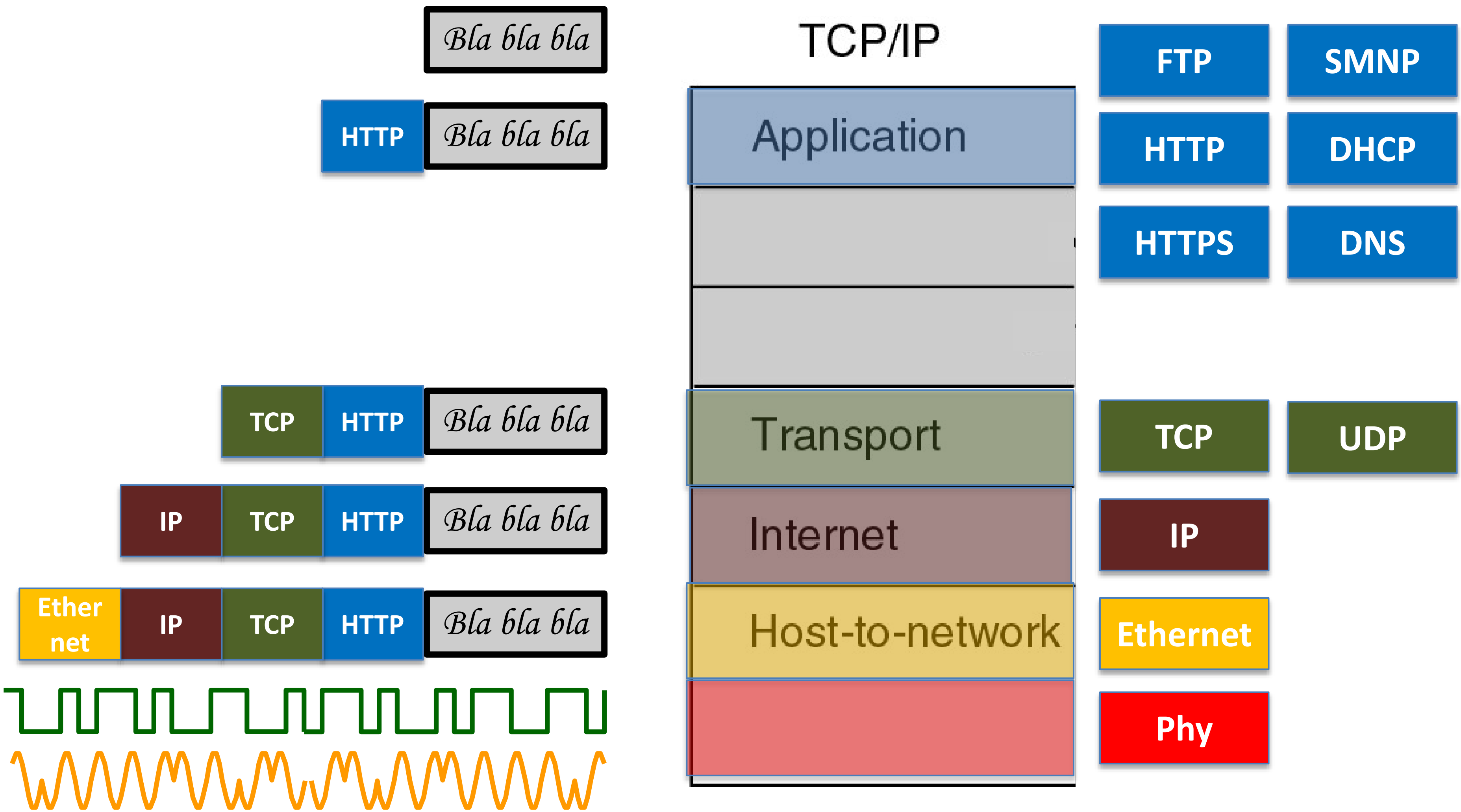
Layers, Services, Interfaces, Entities, Protocols, Models, *etc.*

- (Communication) Rules or agreements between layers are called **protocols.**
- The conceptual network is divided into **layers** to reduce design complexity.
- A layer (n) provides **a service** to the layer above it (n+1) and an **interface** defines the services and operations between them.
- An active element (either software process of a hardware element) in a layer is called an **entity.**

OSI v TCP/IP reference models



OSI v TCP/IP reference models



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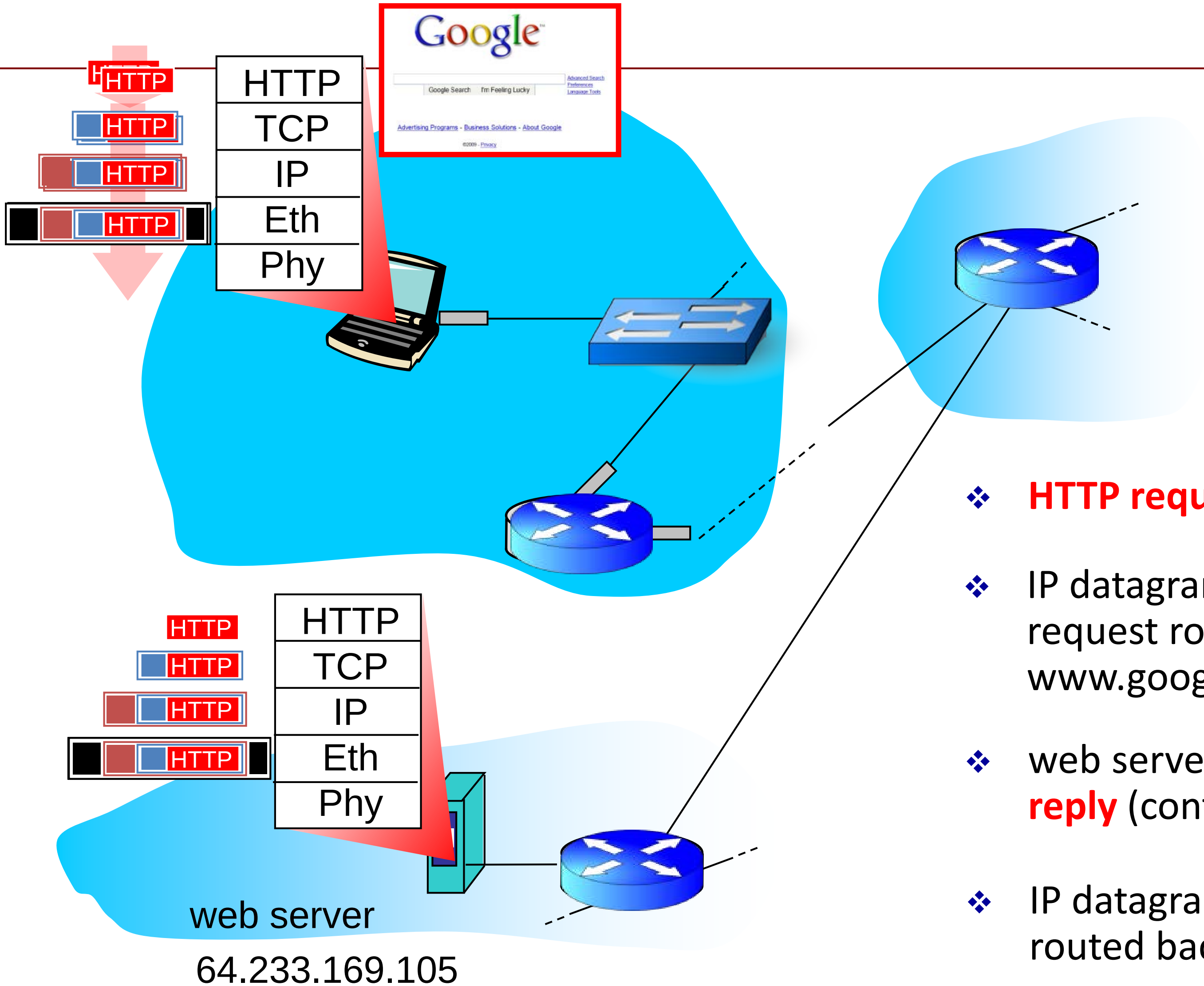
A day in the life… HTTP request/reply

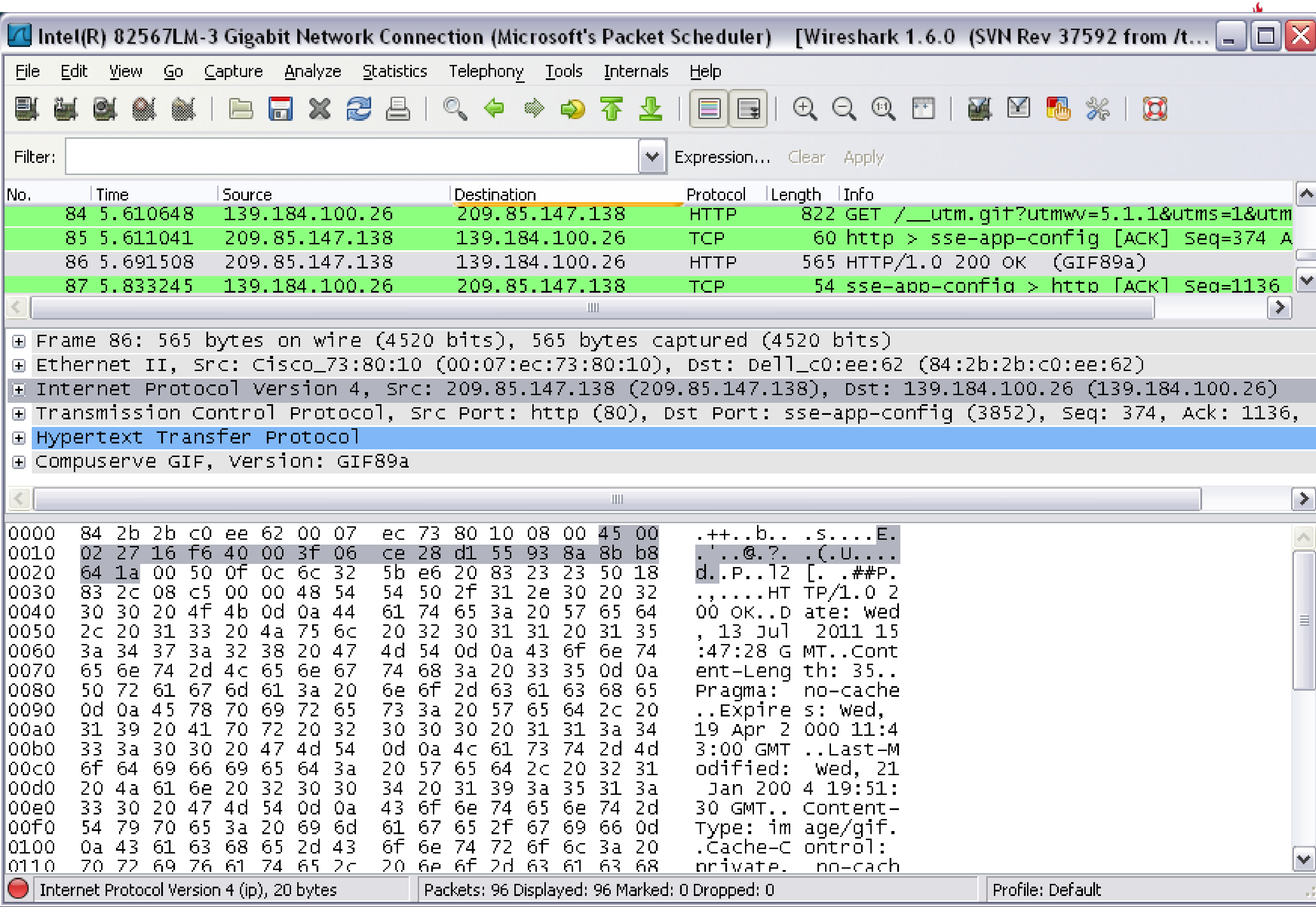


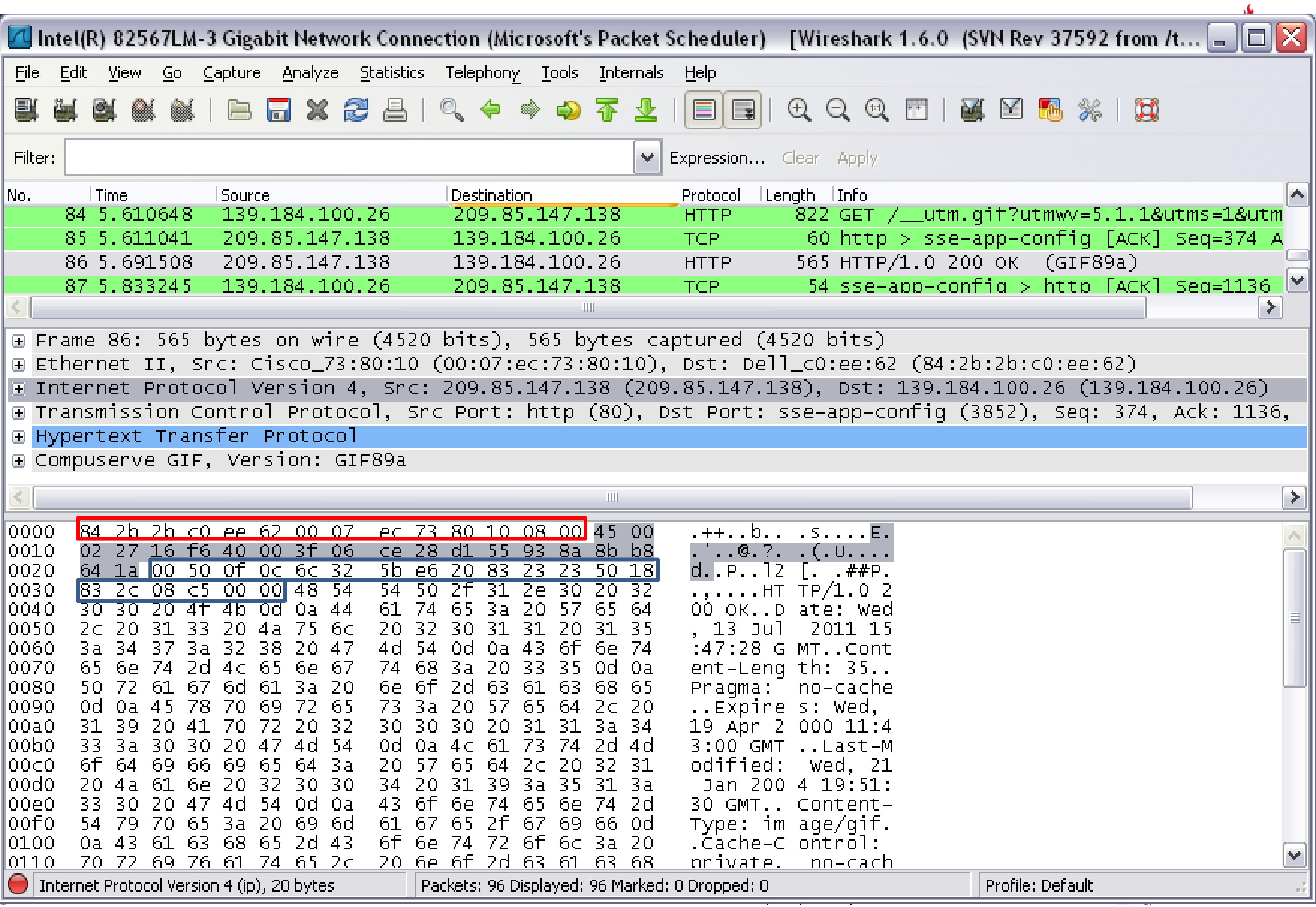
- **HTTP request** sent into TCP socket
- IP datagram containing HTTP request routed to www.google.com
- web server responds with **HTTP reply** (containing web page)
- IP datagram containing HTTP reply routed back to client

What does it really look like?

0000	10000100	00101011	00101011	11000000	11101110	01100010	00000000	00000111
0008	11101100	01110011	10000000	00010000	00001000	00000000	01000101	00000000
0010	00000010	00100111	00010110	11110110	01000000	00000000	00111111	00000110
0018	11001110	00101000	11010001	01010101	10010011	10001010	10001011	10111000
0020	01100100	00011010	00000000	01010000	00001111	00001100	01101100	00110010
0028	01011011	11100110	00100000	10000011	00100011	00100011	01010000	00011000
0030	10000011	00101100	00001000	11000101	00000000	00000000	01001000	01010100
0038	01010100	01010000	00101111	00110001	00101110	00110000	00100000	00110010
0040	00110000	00110000	00100000	01001111	01001011	00001101	00001010	01000100
0048	01100001	01110100	01100101	00111010	00100000	01010111	01100101	01100100
0050	00101100	00100000	00110001	00110011	00100000	01001010	01110101	01101100
0058	00100000	00110010	00110000	00110001	00110001	00100000	00110001	00110101
0060	00111010	00110100	00110111	00111010	00110010	00111000	00100000	01000111
0068	01001101	01010100	00001101	00001010	01000011	01101111	01101110	01110100
0070	01100101	01101110	01110100	00101101	01001100	01100101	01101110	01100111
0078	01110100	01101000	00111010	00100000	00110011	00110101	00001101	00001010
0080	01010000	01110010	01100001	01100111	01101101	01100001	00111010	00100000
0088	01101110	01101111	00101101	01100011	01100001	01100011	01101000	01100101
0090	00001101	00001010	01000101	01111000	01110000	01101001	01110010	01100101
0098	01110011	00111010	00100000	01010111	01100101	01100100	00101100	00100000
00a0	00110001	00111001	00100000	01000001	01110000	01110010	00100000	00110010
00a8	00110000	00110000	00110000	00100000	00110001	00110001	00111010	00110100
00b0	00110011	00111010	00110000	00110000	00100000	01000111	01001101	01010100
00b8	00001101	00001010	01001100	01100001	01110011	01110100	00101101	01001101
00c0	01101111	01100100	01101001	01100110	01101001	01100101	01100100	00111010
00c8	00100000	01010111	01100101	01100100	00101100	00100000	00110010	00110001
00d0	00100000	01001010	01100001	01101110	00100000	00110010	00110000	00110000
00d8	00110100	00100000	00110001	00111001	00111010	00110101	00110001	00111010
00e0	00110011	00110000	00100000	01000111	01001101	01010100	00001101	00001010
00e8	01000011	01101111	01101110	01110100	01100101	01101110	01110100	00101101
00f0	01010100	01111001	01110000	01100101	00111010	00100000	01101001	01101101
00f8	01100001	01100111	01100101	00101111	01100111	01101001	01100110	00001101
0100	00001010	01000011	01100001	01100011	01101000	01100101	00101101	01000011
0108	01101111	01101110	01110100	01110010	01101111	01101100	00111010	00100000
0110	01110000	01110010	01101001	01110110	01100001	01110100	01100101	00101100
0118	00100000	01101110	01101111	00101101	01100011	01100001	01100011	01101000
0120	01100101	00101100	00100000	01101110	01101111	00101101	01100011	01100001
0128	01100011	01101000	01100101	00111101	01010011	01100101	01110100	00101101
0130	01000011	01101111	01101111	01101011	01101001	01100101	00101100	00100000
0138	01110000	01110010	01101111	01111000	01111001	00101101	01110010	01100101

0000	84	2b	2b	c0	ee	62	00	07	ec	73	80	10	08	00	45	00
0010	02	27	16	f6	40	00	3f	06	ce	28	d1	55	93	8a	8b	b8
0020	64	1a	00	50	0f	0c	6c	32	5b	e6	20	83	23	23	50	18
0030	83	2c	08	c5	00	00	48	54	54	50	2f	31	2e	30	20	32
0040	30	30	20	4f	4b	0d	0a	44	61	74	65	3a	20	57	65	64
0050	2c	20	31	33	20	4a	75	6c	20	32	30	31	31	20	31	35
0060	3a	34	37	3a	32	38	20	47	4d	54	0d	0a	43	6f	6e	74
0070	65	6e	74	2d	4c	65	6e	67	74	68	3a	20	33	35	0d	0a
0080	50	72	61	67	6d	61	3a	20	6e	6f	2d	63	61	63	68	65
0090	0d	0a	45	78	70	69	72	65	73	3a	20	57	65	64	2c	20
00a0	31	39	20	41	70	72	20	32	30	30	30	20	31	31	3a	34
00b0	33	3a	30	30	20	47	4d	54	0d	0a	4c	61	73	74	2d	4d
00c0	6f	64	69	66	69	65	64	3a	20	57	65	64	2c	20	32	31
00d0	20	4a	61	6e	20	32	30	30	34	20	31	39	3a	35	31	3a
00e0	33	30	20	47	4d	54	0d	0a	43	6f	6e	74	65	6e	74	2d
00f0	54	79	70	65	3a	20	69	6d	61	67	65	2f	67	69	66	0d
0100	0a	43	61	63	68	65	2d	43	6f	6e	74	72	6f	6c	3a	20
0110	70	72	69	76	61	74	65	2c	20	6e	6f	2d	63	61	63	68
0120	65	2c	20	6e	6f	2d	63	61	63	68	65	3d	53	65	74	2d
0130	43	6f	6f	6b	69	65	2c	20	70	72	6f	78	79	2d	72	65
0140	76	61	6c	69	64	61	74	65	0d	0a	53	65	72	76	65	72
0150	3a	20	47	46	45	2f	32	2e	30	0d	0a	58	2d	43	61	63
0160	68	65	3a	20	4d	49	53	53	20	66	72	6f	6d	20	70	72
0170	6f	78	79	32	2e	75	73	63	73	2e	73	75	73	78	2e	61
0180	63	2e	75	6b	0d	0a	58	2d	43	61	63	68	65	2d	4c	6f
0190	6f	6b	75	70	3a	20	4d	49	53	53	20	66	72	6f	6d	20
01a0	70	72	6f	78	79	32	2e	75	73	63	73	2e	73	75	73	78
01b0	2e	61	63	2e	75	6b	3a	38	30	38	30	0d	0a	56	69	61
01c0	3a	20	31	2e	30	20	70	72	6f	78	79	32	2e	75	73	63
01d0	73	2e	73	75	73	78	2e	61	63	2e	75	6b	3a	38	30	38
01e0	30	20	28	73	71	75	69	64	2f	32	2e	36	2e	53	54	41
01f0	42	4c	45	32	33	29	0d	0a	43	6f	6e	6e	65	63	74	69
0200	6f	6e	3a	20	6b	65	65	70	2d	61	6c	69	76	65	0d	0a
0210	0d	0a	47	49	46	38	39	61	01	00	01	00	80	ff	00	ff
0220	ff	ff	00	00	00	2c	00	00	00	00	01	00	01	00	00	02
0230	02	44	01	00	3b											


Intel(R) 82567LM-3 Gigabit Network Connection (Microsoft's Packet Scheduler) [Wireshark 1.6.0 (SVN Rev 37592 from /t...
File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help
Filter:
Expression... Clear Apply
No. Time Source Destination Protocol Length Info
84 5.610648 139.184.100.26 209.85.147.138 HTTP 822 GET /__utm.gif?utmwv=5.1.1&utms=1&utm
85 5.611041 209.85.147.138 139.184.100.26 TCP 60 http > sse-app-config [ACK] Seq=374 A
86 5.691508 209.85.147.138 139.184.100.26 HTTP 565 HTTP/1.0 200 OK (GIF89a)
87 5.833245 139.184.100.26 209.85.147.138 TCP 54 sse-app-config > http [ACK] Seq=1136
Frame 86: 565 bytes on wire (4520 bits), 565 bytes captured (4520 bits)
Ethernet II, Src: Cisco_73:80:10 (00:07:ec:73:80:10), Dst: Dell_c0:ee:62 (84:2b:2b:c0:ee:62)
Internet Protocol Version 4, Src: 209.85.147.138 (209.85.147.138), Dst: 139.184.100.26 (139.184.100.26)
Transmission Control Protocol, Src Port: http (80), Dst Port: sse-app-config (3852), Seq: 374, Ack: 1136,
Hypertext Transfer Protocol
Compuserve GIF, Version: GIF89a
0000 84 2b 2b c0 ee 62 00 07 ec 73 80 10 08 00 45 00 .++..b.. .s....E.
0010 02 27 16 f6 40 00 3f 06 ce 28 d1 55 93 8a 8b b8 .'..@.?. .(.U....
0020 64 1a 00 50 0f 0c 6c 32 5b e6 20 83 23 23 50 18 d..P..l2 [. .##P.
0030 83 2c 08 c5 00 00 48 54 54 50 2f 31 2e 30 20 32 .,....HT TP/1.0 2
0040 30 30 20 4f 4b 0d 0a 44 61 74 65 3a 20 57 65 64 00 OK..D ate: Wed
0050 2c 20 31 33 20 4a 75 6c 20 32 30 31 31 20 31 35 , 13 Jul 2011 15
0060 3a 34 37 3a 32 38 20 47 4d 54 0d 0a 43 6f 6e 74 :47:28 G MT..Cont
0070 65 6e 74 2d 4c 65 6e 67 74 68 3a 20 33 35 0d 0a ent-Leng th: 35..
0080 50 72 61 67 6d 61 3a 20 6e 6f 2d 63 61 63 68 65 Pragma: no-cache
0090 0d 0a 45 78 70 69 72 65 73 3a 20 57 65 64 2c 20 ..Expire s: Wed,
00a0 31 39 20 41 70 72 20 32 30 30 30 20 31 31 3a 34 19 Apr 2 000 11:4
00b0 33 3a 30 30 20 47 4d 54 0d 0a 4c 61 73 74 2d 4d 3:00 GMT ..Last-M
00c0 6f 64 69 66 69 65 64 3a 20 57 65 64 2c 20 32 31 odified: Wed, 21
00d0 20 4a 61 6e 20 32 30 30 34 20 31 39 3a 35 31 3a Jan 200 4 19:51:
00e0 33 30 20 47 4d 54 0d 0a 43 6f 6e 74 65 6e 74 2d 30 GMT.. Content-
00f0 54 79 70 65 3a 20 69 6d 61 67 65 2f 67 69 66 0d Type: im age/gif.
0100 0a 43 61 63 68 65 2d 43 6f 6e 74 72 6f 6c 3a 20 .Cache-C ontrol:
Internet Protocol Version 4 (ip), 20 bytes
Packets: 96 Displayed: 96 Marked: 0 Dropped: 0
Profile: Default



Intel(R) 82567LM-3 Gigabit Network Connection (Microsoft's Packet Scheduler) [Wireshark 1.6.0 (SVN Rev 37592 from /t...
File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help
Filter:
Expression... Clear Apply
No. Time Source Destination Protocol Length Info
84 5.610648 139.184.100.26 209.85.147.138 HTTP 822 GET /__utm.gif?utmwv=5.1.1&utms=1&utm
85 5.611041 209.85.147.138 139.184.100.26 TCP 60 http > sse-app-config [ACK] Seq=374 A
86 5.691508 209.85.147.138 139.184.100.26 HTTP 565 HTTP/1.0 200 OK (GIF89a)
87 5.833245 139.184.100.26 209.85.147.138 TCP 54 sse-app-config > http [ACK] Seq=1136
Frame 86: 565 bytes on wire (4520 bits), 565 bytes captured (4520 bits)
Ethernet II, Src: Cisco_73:80:10 (00:07:ec:73:80:10), Dst: Dell_c0:ee:62 (84:2b:2b:c0:ee:62)
Internet Protocol Version 4, Src: 209.85.147.138 (209.85.147.138), Dst: 139.184.100.26 (139.184.100.26)
Transmission Control Protocol, Src Port: http (80), Dst Port: sse-app-config (3852), Seq: 374, Ack: 1136,
Hypertext Transfer Protocol
Compuserve GIF, Version: GIF89a
0000 84 2b 2b c0 ee 62 00 07 ec 73 80 10 08 00 45 00 .++..b.. .s....E.
0010 02 27 16 f6 40 00 3f 06 ce 28 d1 55 93 8a 8b b8 .'..@.?. .(.U....
0020 64 1a 00 50 0f 0c 6c 32 5b e6 20 83 23 23 50 18 d..P..l2 [. .##P.
0030 83 2c 08 c5 00 00 48 54 54 50 2f 31 2e 30 20 32 .,....HT TP/1.0 2
0040 30 30 20 4f 4b 0d 0a 44 61 74 65 3a 20 57 65 64 00 OK..D ate: Wed
0050 2c 20 31 33 20 4a 75 6c 20 32 30 31 31 20 31 35 , 13 Jul 2011 15
0060 3a 34 37 3a 32 38 20 47 4d 54 0d 0a 43 6f 6e 74 :47:28 G MT..Cont
0070 65 6e 74 2d 4c 65 6e 67 74 68 3a 20 33 35 0d 0a ent-Leng th: 35..
0080 50 72 61 67 6d 61 3a 20 6e 6f 2d 63 61 63 68 65 Pragma: no-cache
0090 0d 0a 45 78 70 69 72 65 73 3a 20 57 65 64 2c 20 ..Expire s: Wed,
00a0 31 39 20 41 70 72 20 32 30 30 30 20 31 31 3a 34 19 Apr 2 000 11:4
00b0 33 3a 30 30 20 47 4d 54 0d 0a 4c 61 73 74 2d 4d 3:00 GMT ..Last-M
00c0 6f 64 69 66 69 65 64 3a 20 57 65 64 2c 20 32 31 odified: Wed, 21
00d0 20 4a 61 6e 20 32 30 30 34 20 31 39 3a 35 31 3a Jan 200 4 19:51:
00e0 33 30 20 47 4d 54 0d 0a 43 6f 6e 74 65 6e 74 2d 30 GMT.. Content-
00f0 54 79 70 65 3a 20 69 6d 61 67 65 2f 67 69 66 0d Type: im age/gif.
0100 0a 43 61 63 68 65 2d 43 6f 6e 74 72 6f 6c 3a 20 .Cache-C ontrol:
0110 70 72 69 76 61 74 65 2c 20 6e 6f 2d 63 61 63 68 private, no-cach
Internet Protocol Version 4 (ip), 20 bytes
Packets: 96 Displayed: 96 Marked: 0 Dropped: 0
Profile: Default

Network Security

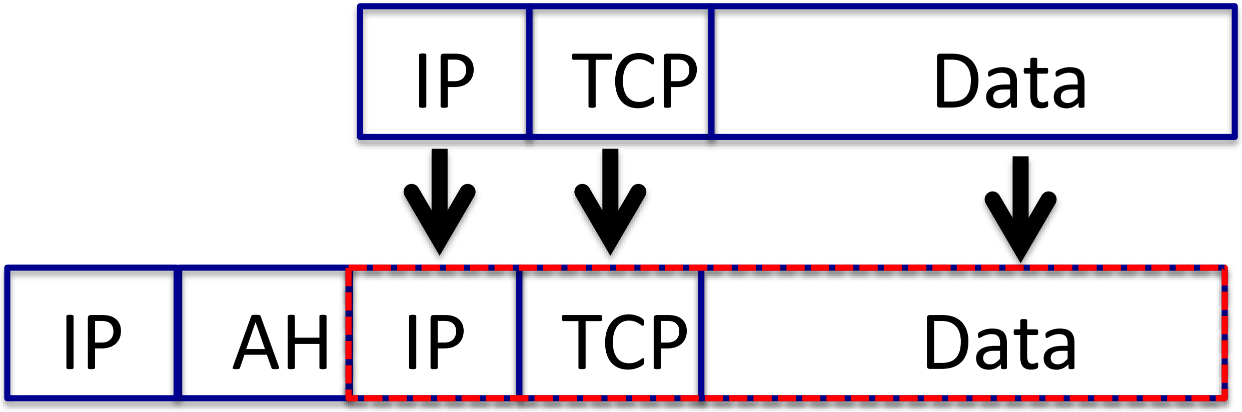
- Introduction to Networks
- Network Security
- Encryption
- Further Topics
- Wrap up

Where is network security?

- *Physical Layer*: encapsulating transmission lines or using fibre optics
- *Data Link Layer*: encrypting packets from point to point
- *Network Layer*: IPsec, firewalls
- *Transport Layer*: entire connections being encrypted
- *Intermediate layers*: Secure Socket Layer

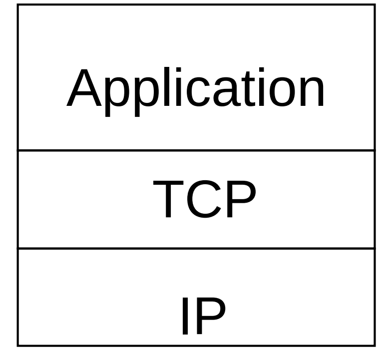
IP Security

- Where to put the security?
  - **Authentication Header** (AH). It provides integrity and anti-replay security, but not secrecy.
  - **Encapsulating Security Payload** (ESP) better option can manage secrecy.

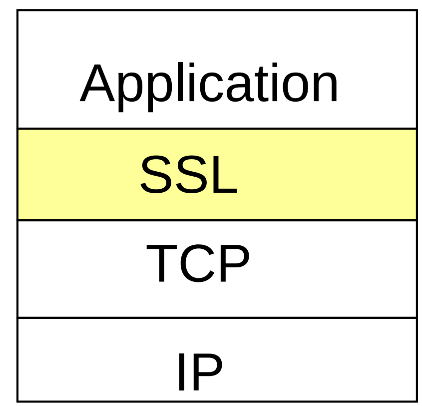


SSL: Secure Sockets Layer

- Widely deployed security protocol
- Variation -TLS: transport layer security, RFC 2246
- provides
  - *confidentiality*
  - *integrity*
  - *authentication*

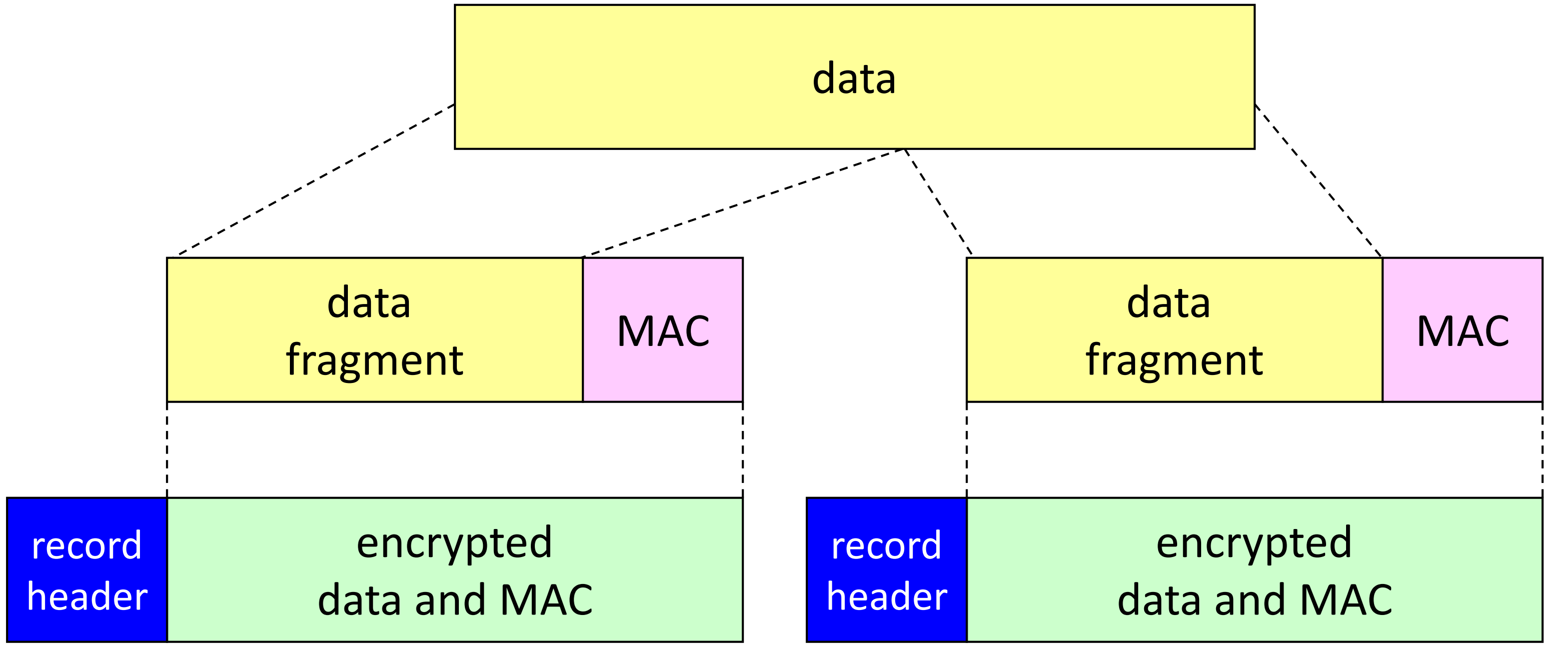


Normal Application



Application w/ SSL

SSL Record Protocol

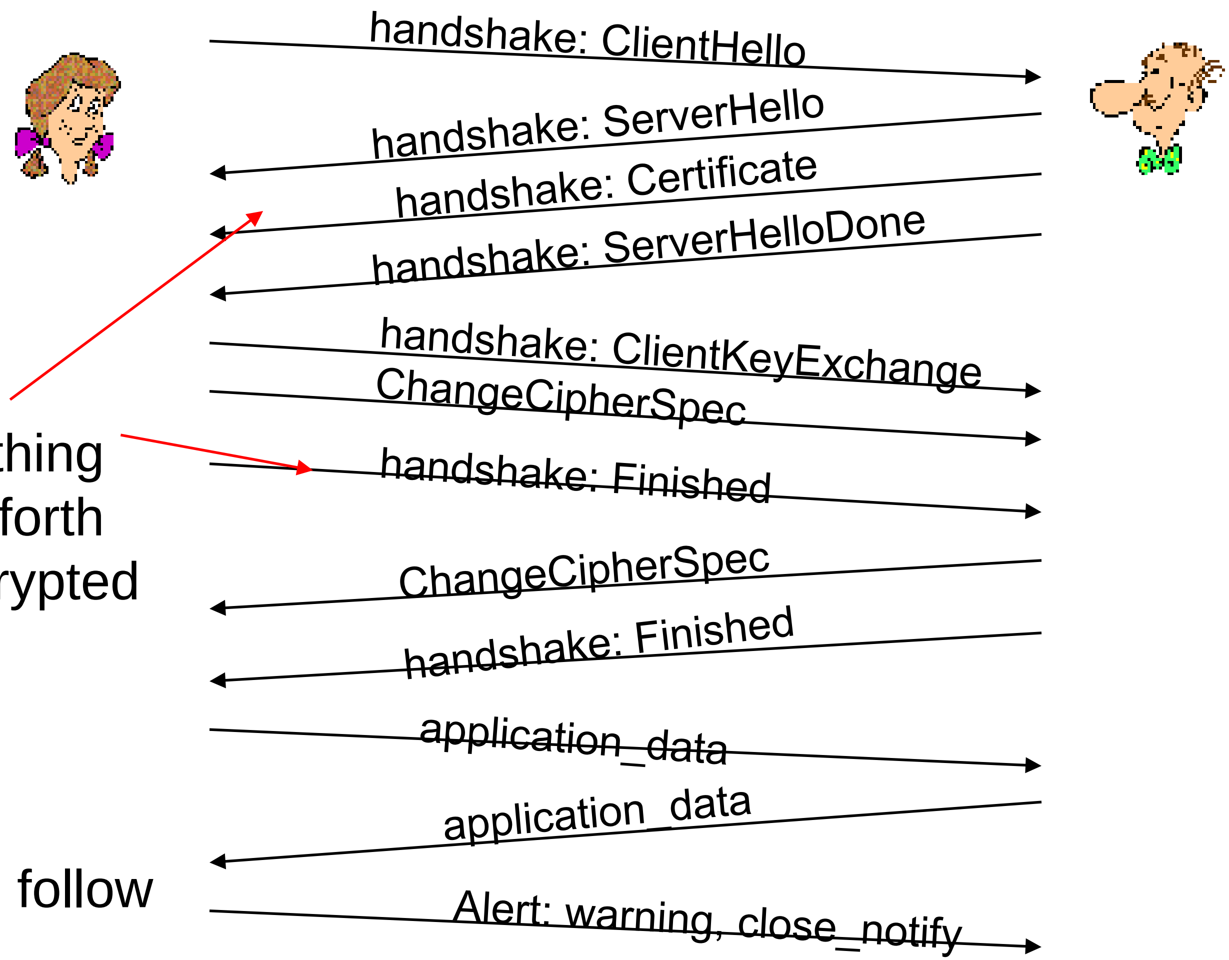


record header: content type; version; length

MAC: includes sequence number, MAC key M_x

fragment: each SSL fragment 2^{14} bytes (~16 Kbytes)

Real Connection



Everything henceforth is encrypted

TCP Fin follow

Network Security (summary)

Basic techniques…...

- cryptography (symmetric and public)
- message integrity
- end-point authentication

…. used in many different security scenarios

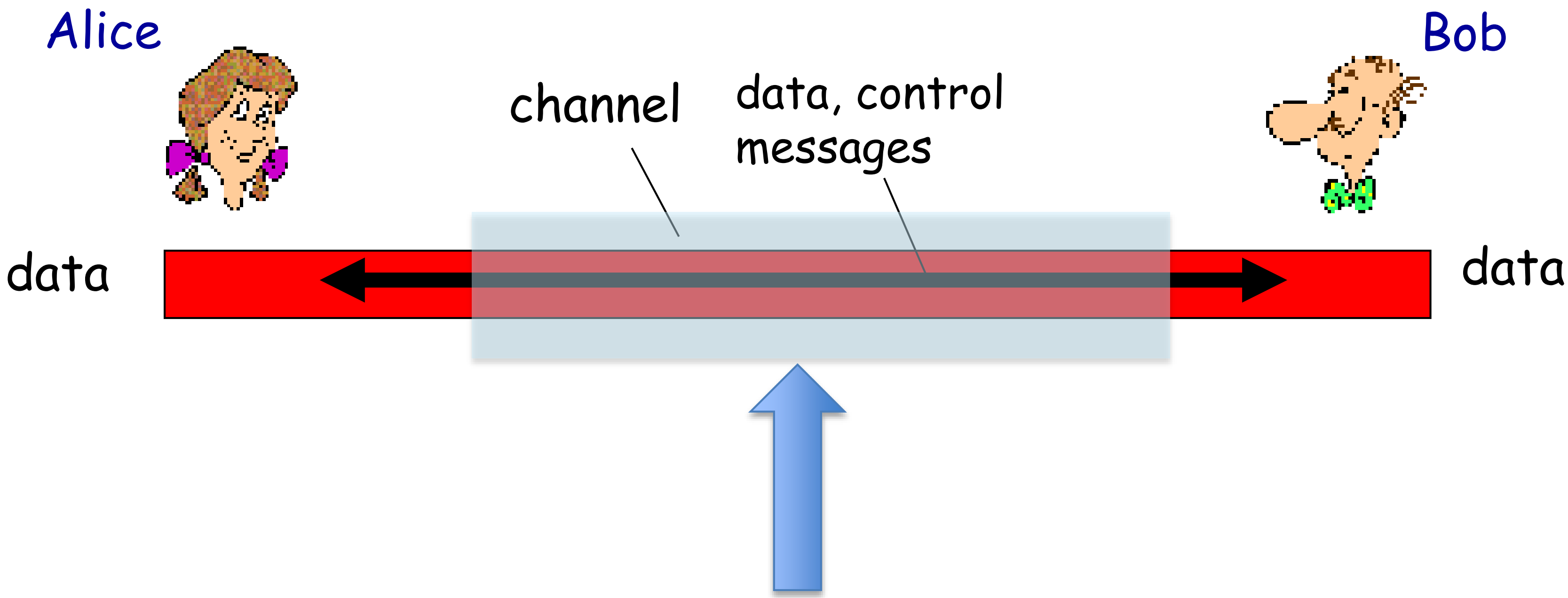
- secure transport (SSL)
- IP sec

And many many others … secure mail, 802.11, firewalls, IDS, etc.

Network Security

- Introduction to Networks
- Network Security
- Encryption
- Further Topics
- Wrap up

Secure Communication over Insecure Medium



Use Cryptography!

Cryptography as a Security Tool

- Broadest security tool available
  - Internal to a given computer, source and destination of messages can be known and protected
    - OS creates, manages, protects process IDs, communication ports
  - Source and destination of messages on network cannot be trusted without cryptography
    - Local network – IP address?
      - Consider unauthorized host added
    - WAN / Internet – how to establish authenticity
      - Not via IP address

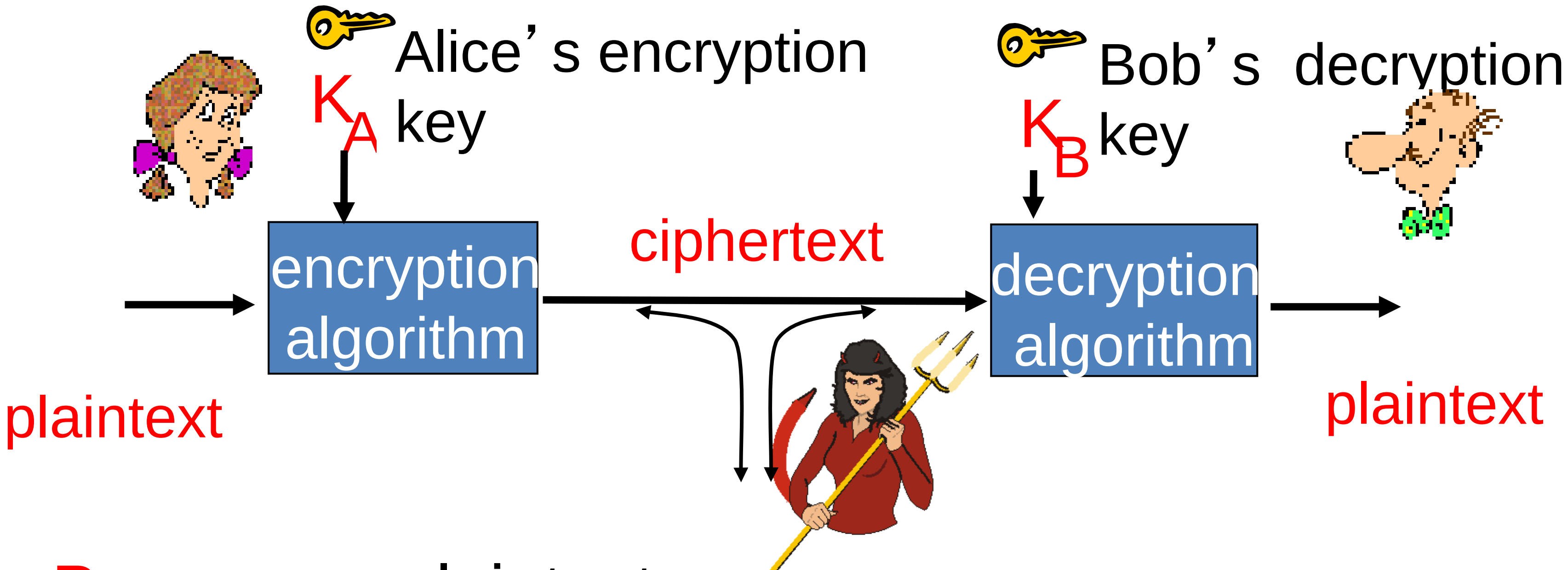
Encryption

- Constrains the set of possible receivers of a message
- Encryption algorithm consists of
  - Set K of keys
  - Set M of Messages
  - Set C of ciphertexts (encrypted messages)
  - A function $E : K \rightarrow (M \rightarrow C)$. That is, for each $k \in K$, E_k is a function for generating ciphertexts from messages
    - Both E and E_k for any k should be efficiently computable functions
  - A function $D : K \rightarrow (C \rightarrow M)$. That is, for each $k \in K$, D_k is a function for generating messages from ciphertexts
    - Both D and D_k for any k should be efficiently computable functions

Encryption (Cont.)

- An encryption algorithm must provide this essential property: Given a ciphertext $c \in C$, a computer can compute m such that $Ek(m) = c$ only if it possesses k
  - Thus, a computer holding k can decrypt ciphertexts to the plaintexts used to produce them, but a computer not holding k cannot decrypt ciphertexts
  - Since ciphertexts are generally exposed (for example, sent on the network), it is important that it be infeasible to derive k from the ciphertexts

The language of cryptography



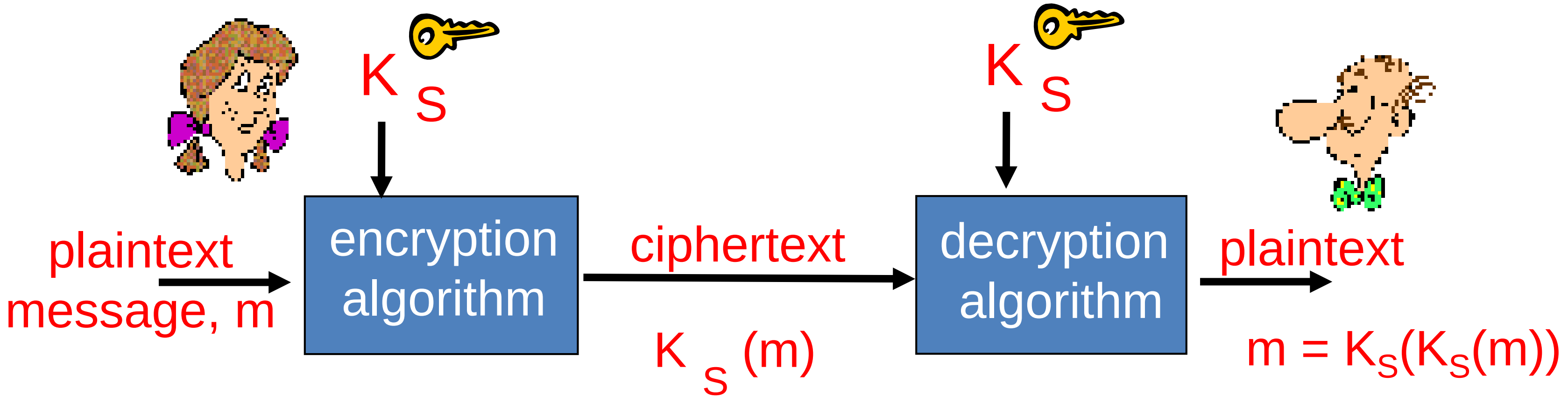
m, P plaintext message

$K_A(m)$, $E(P)$ ciphertext, encrypted with key K_A, E

$m = K_B(K_A(m))$

$D(E(P)) = P$

Symmetric key cryptography



symmetric key crypto: Bob and Alice share same (symmetric) key K_s or one can be derived from the other one.

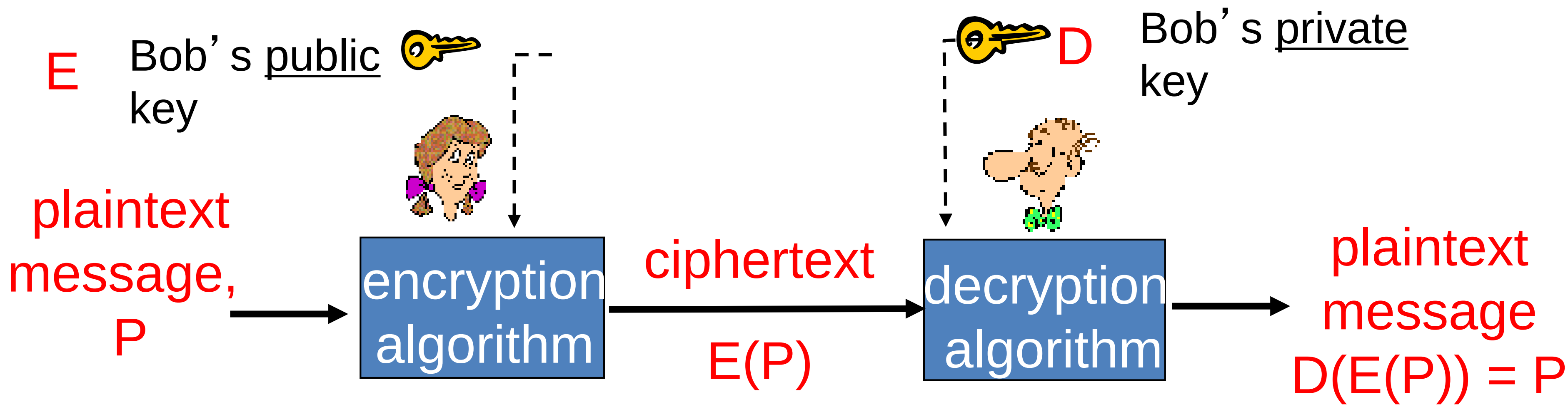
- e.g., key is knowing substitution pattern in mono alphabetic substitution cipher

Q: *how do Bob and Alice agree on key value?*

Symmetric key cryptography

- Historically, distributing the keys has always been *the weakest link* in most cryptosystems.
- No matter how strong a cryptosystem was, if an intruder could steal the key, the system was worthless.
- Cryptologists always took for granted that the encryption key and decryption key were the same (or easily derived from one another).

Public key cryptography



Requirements:

1. D(E(P)) = P
2. It is exceedingly difficult to deduce D from E.
3. E cannot be broken by a chosen plaintext attack.

Decode the following message

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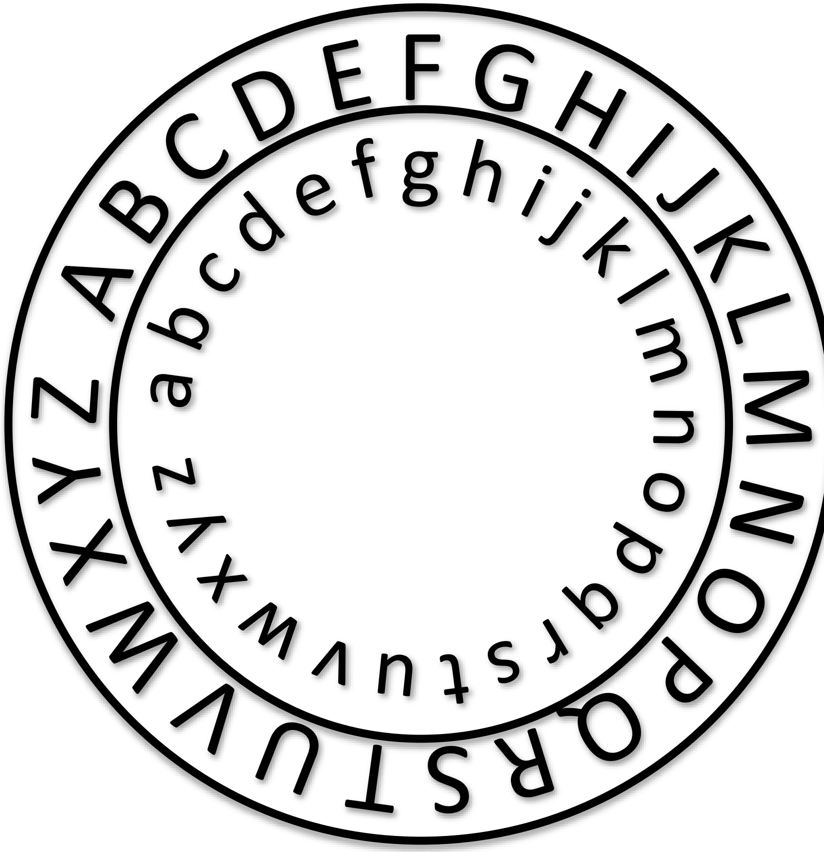
Xibu’t jo b obnf?

Uibu xijdj xf dbmm b sptf

Cz boz puifs obnf xpvme tnfmm bt txffu.



Decode the following message

wnwabelmehahlyrleeamilanslt

tecanatlssthrymiaoihwooelsr

naestwlsoatcehiswm

Decode the following message

```
wnwabelme
hahlyrlee
amilanslt
tecanatls
sthrymiao
ihwooelsr
naestwlso
atcehiswm
```

Simple encryption schemes

- Substitution cipher
  - substitute one letter for another

```
plaintext:   abcdefghijklmnopqrstuvwxyz
ciphertext:  bcdefghijklmnopqrstuvwxyza
```

DOWNSIDE? STATISTICAL ANALYSIS

Simple encryption schemes

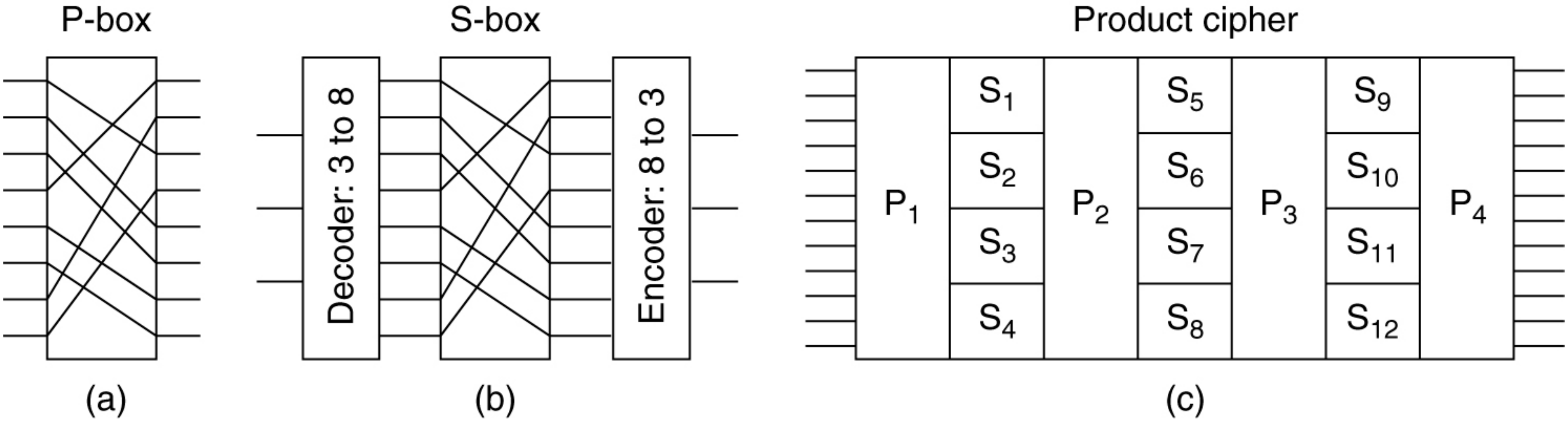
- Substitution cipher
  - substitute one letter for another
- Transposition cipher
  - keep the same characters but change the order

```
plaintext:  abcdefghijklmnopqrstuvwxyz

ciphertext: bcdefghijaklmnopzqrstuvwxy
```

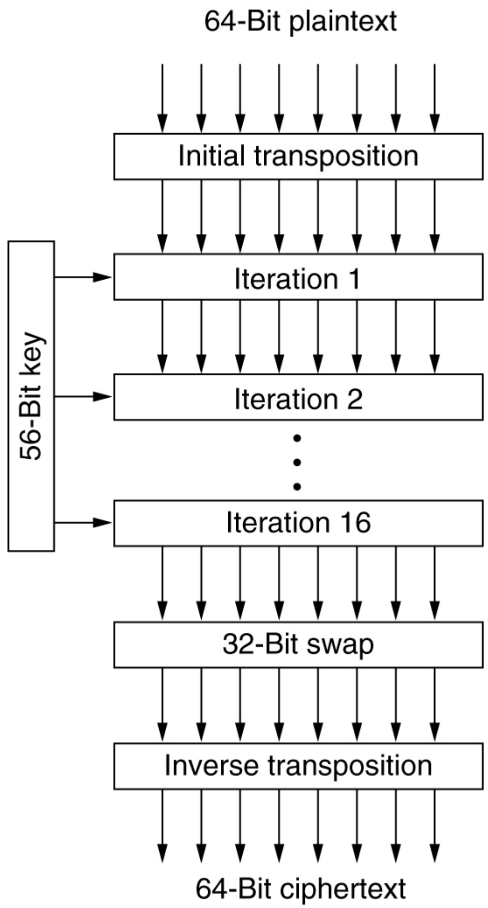
Not-so-simple schemes

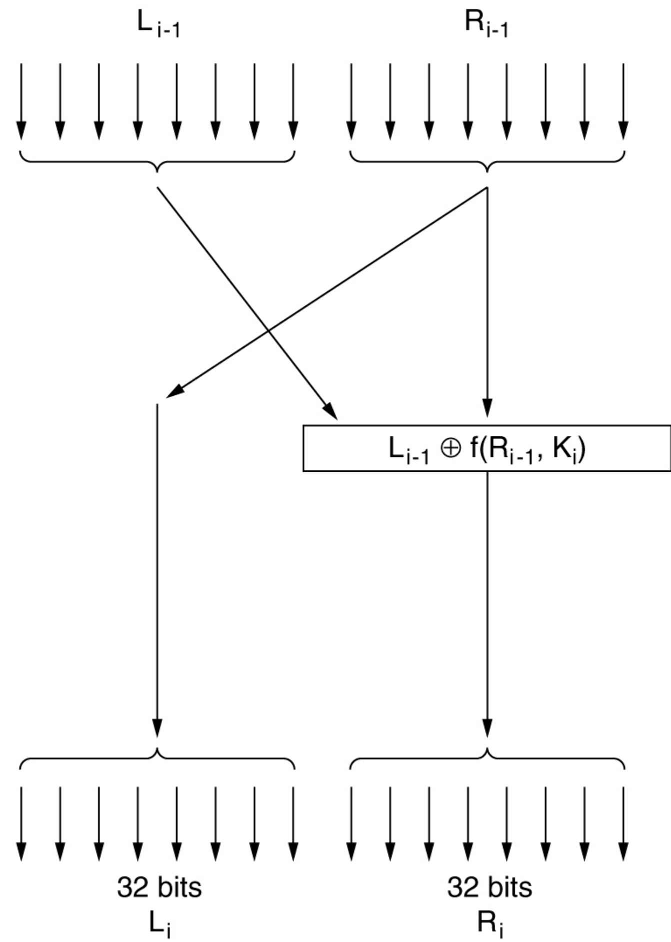
- Product ciphers
  - Combine Permutations, Substitutions into several steps



Not-so-simple schemes

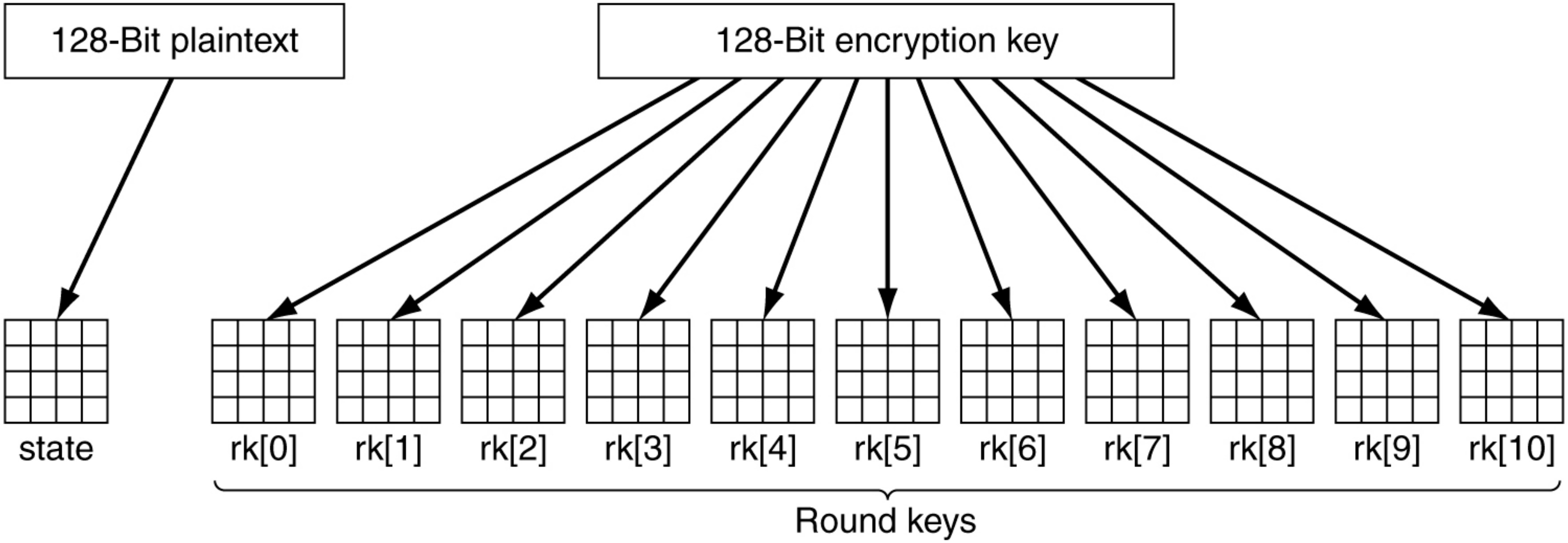
- DES: Data Encryption Standard
  - Break text into 64 bit blocks
  - Process in 19 iterations transpositions and combinations





Not-so-simple schemes

- AES: Advanced Encryption Standard
  - Based on Galois Theory …
  - Substitutions, permutations, multiple rounds
  - 128 bit encryption keys



Monoaphabetic substitution cipher

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EST 1894

- Despite their huge complexity, DES and AES are basically a *monoalphabetic* substitution ciphers using big characters:

message m (t):

What’s in a name that which we call a rose …

encryption algorithm

ciphertext:

AS1234asd5t1234 1234s aas21df32423

Halfway through a message

message m(t+q):

What’s in a name that which we call a rose …

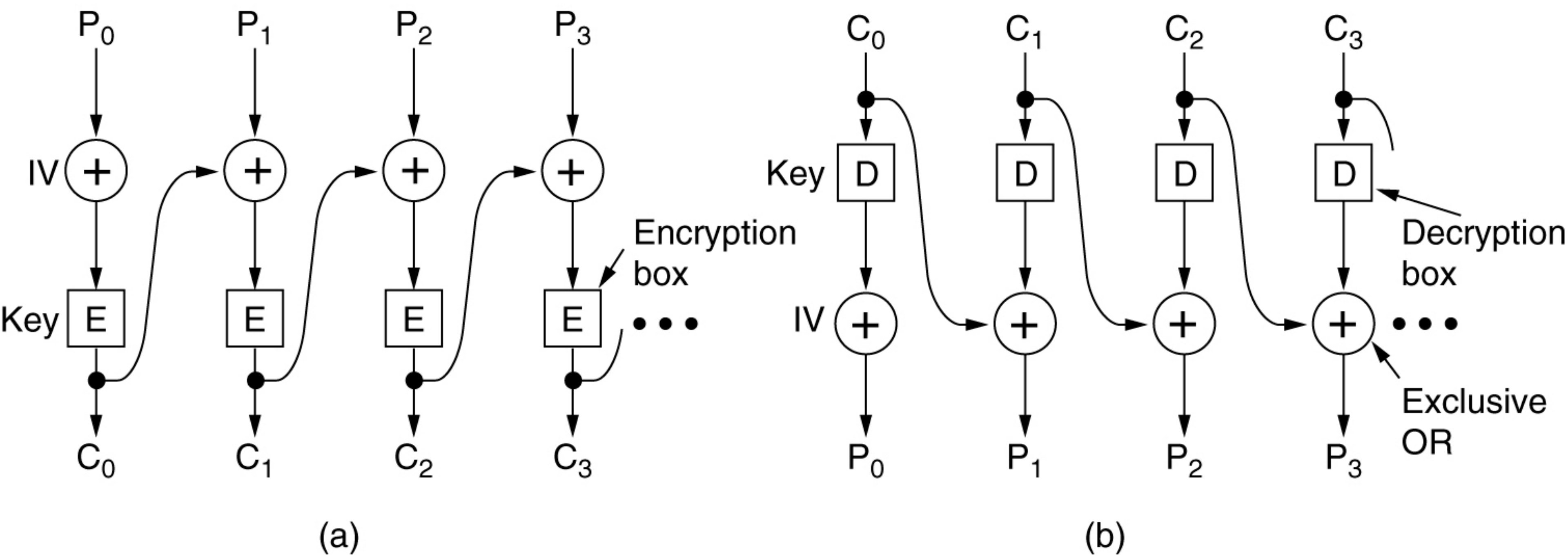
encryption algorithm

ciphertext:

AS1234asd5t1234 1234s aas21df32423

Cipher block chaining

- To make things more difficult change the way in which data is encrypted by chaining the messages:



Cryptography principles

- Sharing keys is the most difficult and vulnerable part of any encryption process
  - One time pads are unbreakable
- Messages must contain some redundancy
  - Random sequences should not correspond to a valid message
- Some method is needed to foil replay attacks
  - Each message should be fresh to prevent the replay of old messages

Cryptanalyst's perspective

- Ciphertext-only problem: no access to plaintext.
- Known plaintext problem: some matched plaintext available.
- Chosen plaintext problem: the Cryptanalyst has the ability to encrypt pieces of plaintext.
- Codes should be unbreakable against encryption of any amount of plaintext.
- Kerckhoff's principle: Algorithms are public, only keys are secret. Security by obscurity ***never works***.

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Multiple cores and processors

- The module has dealt with issues of single CPU systems.
- Most modern systems use multiple cores (implemented on a single chip, each core appearing as a separate processor to the OS), or multiple CPU systems.
- The advantages are the speed-up that this provides in computation.
- But they require more complex process scheduling algorithms, that bring issues about load balancing, and data dependencies.

Threads and synchronisation

- Related to the issues of parallelism and multiple cores, is the issue of multithreading: execution of multiple threads to improve resource sharing and responsiveness.
- But they also bring issues that programmers need to deal with, related to thread synchronisation and scheduling.
- There will be more coverage of these issues in a stage three elective module called “Advanced programming: concurrency”

File systems and storage

- There are more issues to consider with file systems and storage for delivering reliability guarantees to users. We briefly looked at RAID.
- File systems used for cloud storage and large data-stores may have different requirements compared with traditional file systems (e.g. have a look at the Google File System)
- There will be more coverage of cloud computing and associated file systems requirements in the stage 3 elective “Cloud Computing” module.

Security

- We started discussing *some* of the security issues and only briefly discussed possible countermeasures.
- There are a lot more threats and possible countermeasures (e.g. cryptography, application security etc.)
- Some will be discussed in the Computer Networks module in stage 2. A more detailed coverage is given in the “Information Security Fundamentals” module –a stage 3 elective.

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Module wrap-up

- Wrap up will be conducted during the live session.
- If you have questions, about any topic in the module prepare them and ask during the live session, or send them in advance so that we can prepare answers for them.